

CSCE 670 INFORMATION RETRIEVAL · MODULE 13 · LECTURE NOTES

Claiming Relevance Honestly

What a retrieval evaluation establishes.

MODULE OUTCOMES

- State what a retrieval result establishes.
- Identify the standard overclaims.
- Write a defensible system claim.
- Place this course relative to CSCE 638 and CSCE 676.
- State a proportionate practice for building search.

1 What a result establishes

A METRIC IMPROVEMENT IS A STATEMENT ABOUT A JUDGEMENT SET, A METRIC, AND A QUERY SAMPLE

It establishes that on these particular queries, under these particular judgements, by this particular metric, the new system scored higher — four clauses, and the judgements are the one people routinely treat as objective when they are the most contestable.

And the judgements encode a user model (Module 05 §3's metric-as-user-model point) — so a different metric can reverse the ranking of two systems, which is not rare and is worth checking before reporting a single number.

It does not establish that users will be better served, because relevance judgements are a proxy for an information need that the judge did not actually have (Module 01 §1's governing fact) — which is the gap the whole course has been working within.

And on an older test collection, the improvement may partly reflect pooling (Module 05 §1) — so a gain from a method unlike the pool's original builders is systematically understated, and a gain from a similar method is flattered, which is a bias in a specific and predictable direction.

The claim	The correction
'Our system is better.'	On which queries, by which metric, against which baseline? (Module 05 §4.)
'NDCG improved by 3%.'	Mean or median? And how many queries got worse? (Module 05 §4's per-query distribution.)
'Semantic search beats keyword search.'	In domain. Out of domain it loses (Module 07 §3), and on identifier queries it loses badly.

The claim	The correction
'We beat BM25.'	Tuned, or library default? (Module 03 §3's callout.) The two are different claims.
'The answer is cited.'	Does the citation actually support the claim? (Module 11 §2.)
'The ranking is neutral.'	It allocates attention (Module 12 §1). Neutral by which criterion? See the note.

The last row is the one with consequences outside the system, and it is worth attacking hardest: **'neutral' is not a property a ranking can have, because any ordering allocates attention and every allocation implements some criterion** — including the criterion of whatever the scoring function happened to optimise. **A system cannot decline to choose**; it can only decline to say which choice it made (Module 12 §3's value judgement).

2 Claims you can defend

- **'On 50 queries written from real information needs, with relevance judgements we made against written criteria, NDCG@10 is 0.61 against a tuned BM25's 0.54; 38 queries improved and 7 got worse.'** Every clause checkable, and the got-worse count is the one that changes how it reads.
- **'First-stage recall at 500 is 0.89; the cross-encoder reranker contributes 0.04 NDCG over that candidate set.'** Stage-attributed (Module 06 §4), which tells a reader which half to invest in.
- **'Median latency 40 ms, 95th percentile 180 ms, at an approximate-search recall of 0.95.'** **Quality and cost reported together** (Module 04 §4, Module 08 §4), which is the only way either is meaningful.
- **'Of 20 sampled answer sentences, 14 were genuinely supported by the passages cited for them.'** A measured attribution rate (Module 11 §2) rather than a claim that the system cites its sources.
- **And: 'we have not evaluated on identifier or exact-match queries, or on any collection other than this one.'** The limitation statement is Project 2's hardest-graded requirement, and the queries-got-worse count is the clause that most changes how a result is read — which is why both are required.

3 The semester

Course	Its proxy	What the proxy omits
CSCE 638	A metric standing in for a quality judgement — BLEU, ROUGE, a judge model.	Factuality, and whether anybody was actually helped (its Module 10 §3).
CSCE 676	A score standing in for a pattern being real — modularity, silhouette, support.	How hard you looked (its Module 11) — the search is invisible in the score.
CSCE 670 (this one)	A relevance label standing in for relevance to a person.	The need itself, the session, and redundancy within a list (Module 01 §3, Module 05 §3).

And the shared discipline is the same in all three: name the proxy, state what it omits, and report the distribution rather than the mean — which is CSCE 633 Module 12's choose-the-metric-from-the-use argument, arriving three separate ways in one semester. The three-proxies framing is the semester's conclusion, and it is worth more than any individual technique in it, because it transfers to every measurement problem.

4 A proportionate practice

- 1 · A tuned BM25 with good preprocessing (Modules 02 and 03) — which gets you most of the way and is cheap, and is where a surprising number of systems should simply stop, having met their requirement.
- 2 · Spelling correction (Module 09 §1) — the highest-value single addition, because its failures are total rather than degraded and a substantial fraction of queries are affected.
- 3 · A judgement set and a working evaluation (Module 05) — before any further improvement, because otherwise you cannot tell whether anything you do next helps.
- 4 · Hybrid retrieval and a cross-encoder reranker (Modules 06 and 07), with the latency measured at the tail and the stages attributed separately.
- 5 · And then the consequences: exposure, the feedback loop, and diversity (Module 12) — which should not wait for scale, since the feedback loop starts with the first deployment. Step 3 before step 4 is the ordering people get wrong: building the neural system before the evaluation means never knowing whether it helped, and that is the commonest way effort is wasted in this field.

WHERE THIS COURSE LEAVES YOU

You can build a search engine: an inverted index with compression and skip pointers, a scoring function you can derive rather than recite, efficient query processing with safe early termination, and a learned or hybrid ranker in two stages.

You can evaluate one honestly — a judgement set you made against criteria you wrote first, a metric chosen from how the results will be used, a properly tuned baseline, and the per-query distribution rather than the mean — which is the half that most systems omit and the half this course spent its centre on.

And you know that a ranking allocates attention rather than merely ordering documents (Module 12), so the question 'who else is affected by this' is one you now ask as a matter of course rather than when prompted.

The closing rule is the program's, unchanged across thirty courses: state what you measured, state what you assumed, and never claim more than you established. In this subject it means naming the judgement set, the metric, and the baseline — because 'relevant' is a relation and the system never saw one side of it (Module 01 §1), which is where the course began and is what every claim in it has to respect.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Armstrong et al. — Improvements that don't add up (free)	§1's problem, measured across a decade of papers — weak baselines producing apparent cumulative progress that was not there.
Lin — The Neural Hype and Comparisons Against Weak Baselines (free)	§1's fourth row — the tuned-baseline argument, stated directly.
Fuhr — Some Common Mistakes In IR Evaluation (free)	§1 and §2 — a short, pointed list, and worth checking your own work against.
Sakai — Statistical Reform in Information Retrieval? (free)	§2's reporting requirements — significance testing, effect sizes, and the per-query distribution.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Take a published retrieval result** and assess it against §1's table.

- 2 **Check whether its baseline was tuned.**
- 3 **Compute two metrics on the same two systems** and find a case where they disagree.
- 4 **Count the queries your system made worse**, and report it.
- 5 **Write the defensible claim** for your Project 2 system, in §2's form.
- 6 **Write its limitation statement.**
- 7 **Name the proxy** in each of the three Semester 10 courses, and what each omits.
- 8 **Audit your own build order** against §4's five steps.
- 9 **Revisit Module 01's last exercise** and compare.
- 10 **Project 2 is now due.** Submit the collection, the thirty need-derived queries, your judgements and criteria, the tuned baseline, your system, the evaluation with latency, and the twenty-worst-query analysis.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What four clauses does a metric improvement require?
- 2 Why can a different metric reverse two systems' order?
- 3 What does a metric improvement not establish?
- 4 How does pooling bias an improvement, and in which direction?
- 5 Give six overclaims and the correction to each.
- 6 Why can a ranking not be neutral?
- 7 Give four defensible claim forms.
- 8 Name the proxy in each Semester 10 course and what it omits.
- 9 Give the five-step build order, and which ordering people get wrong.
- 10 In this course, what does 'never claim more than you established' mean?